

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q18-1060

Griffiss Air Base 100 kV  
March 17, 2020

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1060. The Interconnection Customer proposes a 187.6 MW generating facility interconnecting at the Griffiss Air Base 100 kV substation in the Illinois area. The request was received on September 10, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 348 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,300,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                                  | Contingency                           | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------------------------|---------------------------------------|-------------|--------------|----------|
| Blythewood – Nixon Rd 100 KV Line 1                                 | Blythewood – Nixon Rd 100 KV Line 2   | 143.7       | 153          | 35.2     |
| Nixon Road – Blythewood 100 KV Cir. 2                               | Nixon Road – Blythewood 100 KV Cir. 1 | 143.7       | 153          | 35.2     |
| Griffiss Air Base – Twomile Road 100 KV                             | Ann – Bushland 100 KV                 | 107.2       | 135          | 24.3     |
| Winthrop Substation – Hebo Substation 230 KV                        | Hebo – Barrett 230 KV                 | 101.5       | 1,152        | 5.5      |
| Montpelier Electric Generating Station Substation – Muskogee 500 KV | Lonetree – Sabanea 500 KV             | 101.1       | 4,673        | 10.1     |
| Two Mile D – Voorheesville 100 KV Line                              | Bushland – Ann Substation 100 KV      | 101.0       | 127          | 24.3     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                              | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------|---------------------------------------------|---------------|-----------------------|
| Blythewood – Nixon Rd 100 KV Line 1   | Rebuild line with higher-capacity conductor | \$3,313,000   | 24% (\$781,000)       |
| Nixon Road – Blythewood 100 KV Cir. 2 | Rebuild line with higher-capacity conductor | \$4,799,000   | 24% (\$1,131,000)     |

| Facility                                                            | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------------------------------------|----------------------------|---------------|-----------------------|
| Griffiss Air Base – Two Mile Rd Substation 100 KV line              | Reconductor line           | \$4,038,000   | 33% (\$1,319,000)     |
| Winthrop Substation – Hebo Substation 230 KV                        | Reconductor line           | \$2,890,000   | 2% (\$44,000)         |
| Montpelier Electric Generating Station Substation – Muskogee 500 KV | Reconductor line           | \$41,124,000  | 2% (\$621,000)        |
| Two Mile D – Voorheesville 100 KV Line                              | Reconductor line           | \$1,237,000   | 33% (\$404,000)       |
|                                                                     | Total allocated to project |               | \$4,300,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.